MODULE 3: FUNCTIONAL MATERIALS (ELECTRICAL & MAGNETIC)

Complete Unabridged Course Notes, Derivations, Diagrams & Topic-Wise CAT/FAT Solved Problems

Fully Aligned with Official VIT Faculty Lecture Decks (51 Slides), Department Question Bank, and Past FAT Papers (CO: 3)

Module 3 Detailed Syllabus Breakdown (100% Complete Coverage):

Part A: Electrical Materials: Ohm's law, resistance (R), resistivity (ρ), conductivity (σ), factors affecting resistivity (nature, temperature coefficient α, Matthiessen's rule of impurities), Energy Band Theory for solids (VB, CB, E_g), Conductors vs Semiconductors vs Insulators, Low resistivity conductor materials: Copper (hard-drawn vs annealed Cu properties & applications), Aluminium (electronic config, 61% IACS, ACSR conductors), Semiconductors (intrinsic & extrinsic), Superconductors (Kamerlingh Onnes discovery, T_c, zero resistance, persistent currents, Meissner effect, perfect diamagnetism, critical field $H_{c(T)}$, Type I vs Type II superconductors, High-T_c ceramic YBCO, applications).
Part B: Magnetic Materials: Origin of magnetism (orbital & spin dipole moments, Bohr magneton μ_B), Magnetic terms (B, H, M, χ, μ_r), 5 Types of Magnetic Materials (Diamagnetic, Paramagnetic, Ferromagnetic, Antiferromagnetic, Ferrimagnetic — spin alignments, susceptibility, Curie/Néel temperatures), Hysteresis loop ($B-H$ curve, retentivity, coercivity), Soft vs Hard magnetic materials, Applications in data storage (HDDs, magnetic tapes, MRAM).

PART A: ELECTRICAL MATERIALS

1. Electrical Conduction, Resistance, and Resistivity

1.1 Fundamental Equations of Electrical Conduction

- **Ohm's Law:** At constant physical conditions (temperature, pressure), electric current (I) flowing through a metallic conductor is directly proportional to the applied potential difference (V):

$$V = IR$$

- **Electrical Resistance (R):** The property of a conductor that opposes the flow of electrons, defined by geometry:

$$R = \rho \frac{L}{A}$$

  where L is conductor length (m), A is cross-sectional area (m^2), and ρ is the electrical resistivity (Ω m).
- **Resistivity (ρ):** An intrinsic property quantifying how strongly a material resists electric current:

$$\rho = R\frac{A}{L} \quad \text{(Units: } \Omega \text{ m or } \Omega \text{ cm)}$$

- **Electrical Conductivity (σ):** The inverse of resistivity:

$$\sigma = \frac{1}{\rho} = \frac{L}{RA} \quad \text{(Units: S } m^{-1} \text{ or } \Omega^{-1}m^{-1})$$

  Microscopic formulation: $\sigma = ne\mu$, where n is charge carrier density, $e = 1.602 \times 10^{-19}$ C, and μ is carrier drift mobility.

1.2 Factors Affecting Electrical Resistivity

1. **Nature of the Material:** Governed by the concentration of conduction electrons (n). In metals ($n \approx 10^{28} m^{-3}$), resistivity is extremely low ($10^{-8} \Omega$ m). In insulators ($n \approx 10^{7} m^{-3}$), resistivity is enormous ($10^{12} - 10^{16} \Omega$ m).
2. **Effect of Temperature:**

- **Metals (Positive TCR, $\alpha > 0$):** Thermal agitation causes lattice ions to vibrate with larger amplitude around equilibrium positions. These vibrations (phonons) scatter conduction electrons, decreasing mean free path (λ) and relaxation time (τ). Hence, resistivity increases linearly with temperature:

$$R_T = R_0(1 + \alpha T) \quad \text{or} \quad R_2 = R_1[1 + \alpha_1(T_2 - T_1)]$$

  where α is the Temperature Coefficient of Resistance (K^{-1} or $^\circ\text{C}^{-1}$).
- **Semiconductors and Insulators (Negative TCR, $\alpha < 0$):** Thermal energy breaks covalent bonds, exponentially increasing carrier concentration ($n_i \propto \exp\left(-\frac{E_g}{2}k_B T\right)$). The increase in carrier density overwhelmingly dominates any scattering, causing resistivity to drop exponentially.

3. **Effect of Impurities and Alloying (Matthiessen's Rule):** Foreign solute atoms distort the periodic lattice potential, creating localized scattering centers that impede electron flow. The total resistivity of an impure metal is:

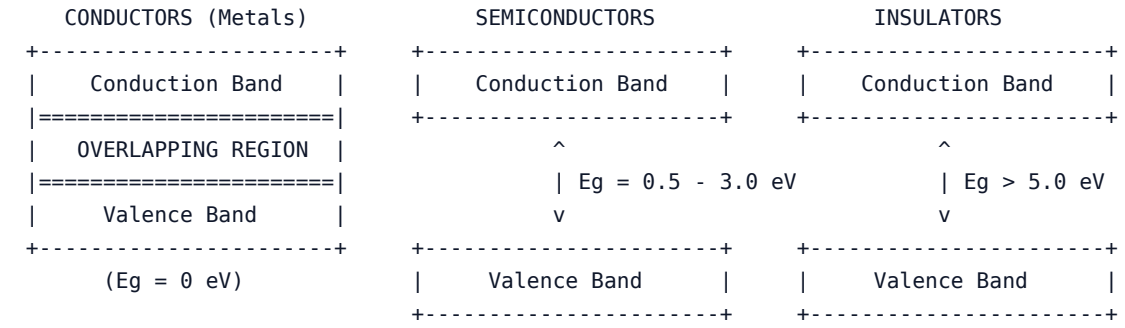
$$\rho_{\text{total}} = \rho_{\text{thermal}(T)} + \rho_{\text{impurity}(c)} + \rho_{\text{defect}}$$

   Adding just 1% impurity can increase copper's electrical resistivity by over $20 - 50\%$.
4. **Mechanical Work / Cold Working:** Cold drawing, rolling, or hammering increases dislocation density and internal grain boundary defects, scattering electrons and increasing resistivity. Annealing relieves internal stress and restores maximum conductivity.

2. Energy Band Theory for Solids

In an isolated atom, electrons occupy discrete atomic orbitals ($1s, 2s, 2p$). When N atoms assemble into a crystal lattice, Pauli's exclusion principle splits closely spaced energy levels into continuous **energy bands**:

- **Valence Band (VB):** The highest energy band filled with valence electrons in ground state.
- **Conduction Band (CB):** The vacant or partially filled band where electrons are free to drift under an electric field.
- **Forbidden Bandgap (E_g):** The forbidden energy gap separating the top of the VB and bottom of the CB.



Parameter	Conductors (Metals)	Semiconductors	Insulators
Bandgap (E_g)	Zero ($E_g = 0$); VB and CB overlap.	Small ($E_g = 0.5 - 3.0$ eV; Si: 1.12 eV, Ge: 0.67 eV).	Very wide ($E_g > 5.0$ eV; Diamond: 5.47 eV).
Resistivity (ρ)	Extremely low ($10^{-8} - 10^{-6}\Omega$ m).	Moderate ($10^{-4} - 10^{4}\Omega$ m).	Extremely high ($10^{10} - 10^{18}\Omega$ m).
Temperature Coefficient (α)	Positive ($\alpha > 0$); resistance increases with T.	Negative ($\alpha < 0$); resistance decreases exponentially with T.	Negative ($\alpha < 0$); negligible conduction until dielectric breakdown.
Charge Carriers	Free conduction electrons only ($n_e \approx 10^{28}\text{m}^{-3}$).	Both electrons (n_e) and positive holes (n_h).	Virtually zero free carriers at normal temperatures.

3. Conductor Materials: Copper and Aluminium

Low-resistivity conducting materials are essential for electrical power generation, transmission, transformer windings, and printed circuit board (PCB) traces.

3.1 Copper (Cu) as a Technical Conductor

- **Atomic & Electronic Properties:** Atomic number 29; configuration $[Ar]3d^{10}4s^1$. The single $4s$ valence electron is loosely held and acts as a completely delocalized conduction electron.
- **High Electrical & Thermal Conductivity:** Second only to silver in electrical conductivity, but vastly more abundant and economical. Standard reference: International Annealed Copper Standard (IACS), where pure annealed copper is defined as 100% IACS ($\rho = 1.7241 \times 10^{-8} \Omega$ m at 20° C).
- **Corrosion Resistance & Solderability:** Forms a protective passivating copper oxide film; easily joined using standard lead-free tin-based solders.

3.2 Hard-Drawn vs. Annealed Copper

Property	Hard-Drawn Copper	Annealed Copper
Manufacturing Method	Drawn through tungsten carbide dies at room temperature without subsequent heat treatment.	Drawn and then heat treated (annealed) in an inert furnace at $500 - 650$° C and slowly cooled.
Microstructure	Elongated, heavily strained fibrous grains with high internal dislocation density.	Equiaxed, stress-free recrystallized grains.
Tensile Strength	**High:** $400 - 450$ MPa ($40 - 45 \text{kg/mm}^2$); resists mechanical sagging.	**Low / Moderate:** $200 - 250$ MPa ($20 - 25 \text{kg/mm}^2$).
Elongation / Ductility	Low elongation ($1 - 4\%$); stiff and rigid.	**High elongation:** $25 - 40\%$; extremely soft, ductile, and flexible.
Electrical Conductivity	$97\% - 98\%$ IACS (dislocations scatter electrons).	**100% IACS** ($\rho = 1.724 \times 10^{-8} \Omega$ m; maximum conductivity).
Primary Applications	Overhead railway catenary wires, transmission spans, busbars, trolley wires.	Transformer windings, motor stator coils, building wiring, flexible cables.

3.3 Aluminium (Al) as an Electrical Conductor

- **Atomic & Electronic Properties:** Configuration $[Ne]3s^2 3p^1$ (trivalent).
- **Density vs. Conductivity Advantage:**
  - Conductivity is 61% IACS (lower than copper), but density is only 2.70g cm^{-3} (one-third that of copper, 8.96g cm^{-3}).
  - **Weight Comparison:** For a transmission line of equal electrical resistance, an aluminium conductor requires a larger diameter ($1.28\times$) but weighs **less than half (48%)** of the equivalent copper conductor!
- **Engineering Limitations:** Lower tensile strength ($150 - 200$ MPa); prone to mechanical creep and wind-induced galloping; forms a tough, insulating dielectric aluminium oxide (Al_2O_3) surface skin that complicates terminal joint connections.
- **The Solution: ACSR Conductor (Aluminium Conductor Steel Reinforced):**
  - Central core consists of 1 to 7 strands of high-tensile galvanized steel wire.
  - Surrounding outer layers consist of multiple helically wound strands of high-purity electrical-grade aluminium (EC grade).
  - The steel core provides immense tensile strength to withstand long spans between towers and wind/ice loading, while the outer aluminium strands carry over 98% of the electrical current due to the AC skin effect.

4. Superconductors and Quantum Levitation

In 1911, Dutch physicist Heike Kamerlingh Onnes discovered that when solid mercury (Hg) is cooled by liquid helium below 4.2 K, its electrical resistance abruptly plunges to absolute zero ($R = 0$).

4.1 Fundamental Characteristics of the Superconducting State

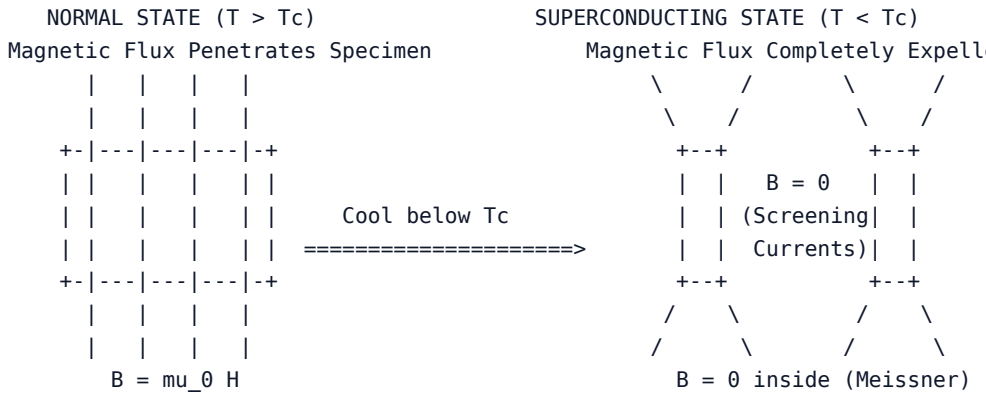
1. **Zero Electrical Resistance ($\rho = 0$):** Below the Critical Temperature (T_c), electrical resistivity is strictly zero ($\rho < 10^{-25} \Omega$ m). An electrical current induced in a superconducting closed ring continues to circulate indefinitely without measurable decay (**persistent currents**).
2. **Critical Temperature (T_c):** The exact thermodynamic transition temperature at which a normal conducting material transitions into the superconducting state.
3. **Critical Magnetic Field (H_c):** If an external magnetic field applied to a superconductor exceeds a threshold value H_c, superconductivity is instantly destroyed and the material returns to its normal resistive state:

$$H_{c(T)} = H_0 \left[1 - \left(\frac{T}{T_c} \right)^2 \right]$$

where H_0 is the critical field at absolute zero ($T = 0$ K).

4.2 The Meissner Effect and Perfect Diamagnetism

Discovered in 1933 by Walther Meissner and Robert Ochsenfeld: "When a superconducting specimen is cooled below its critical temperature T_c in the presence of an applied magnetic field, all internal magnetic flux lines are completely expelled from the interior of the material."



- **Mathematical Formulation:** Inside the bulk superconductor:

$$\boldsymbol{B = \mu_0 (H + M) = 0}$$

Therefore:

$$M = -H \Longrightarrow \boldsymbol{\chi = \frac{M}{H} = -1}$$

- **Magnetic Susceptibility:** $\chi = -1.0$ (Strictly negative unit magnitude).
- **Relative Permeability:** $\mu_r = 1 + \chi = 1 + (-1) = \mathbf{0}$.

Crucial Deduction: Superconductivity is NOT merely "zero electrical resistance"; it is a distinct thermodynamic equilibrium state characterized by **perfect diamagnetism**.

4.3 Type I vs. Type II Superconductors

Parameter	Type I Superconductors (Soft)	Type II Superconductors (Hard)
Critical Fields	Single critical magnetic field H_c.	Two critical magnetic fields: Lower H_{c1} and Upper H_{c2}.

Meissner Effect	Complete, abrupt flux expulsion up to H_c; then sharp transition to normal state.	Complete Meissner effect below H_{c1}; **Vortex/Mixed State** between H_{c1} and H_{c2}.
Critical Field Values	Very low ($H_c \approx 0.01 - 0.1$ Tesla); destroyed by weak magnetic fields.	Extremely high ($H_{c2} > 20 - 100$ Tesla); withstands intense magnetic fields.
Critical Temp (T_c)	Low ($T_c < 10$ K, requires liquid helium cooling).	High (T_c up to $93 - 135$ K in ceramic cuprates; liquid nitrogen cooling).
Material Types	Pure metals (Pb, Hg, Sn, Al, Zn).	Intermetallic alloys and ceramic oxides ($Nb_3Sn, Nb-Ti, YBa_2Cu_3O_7$).
Engineering Utility	Cannot carry high currents; useless for high-field electromagnets.	Used for MRI scanners, Maglev trains, and CERN particle accelerators.

4.4 High-T_c Ceramic Superconductor: $YBa_2Cu_3O_{7-\delta}$ (YBCO)

- Discovered in 1987 by Paul Chu and Maw-Kuen Wu: Yttrium Barium Copper Oxide ($YBa_2Cu_3O_7$).
- **Layered Perovskite Crystal Structure:** Features triple-layer defect perovskite stacking. Superconductivity is confined to two-dimensional copper-oxygen planes (CuO_2), where Cooper pairing of holes occurs.
- **Critical Temperature: $T_c = 93$ K**.
  - **The Liquid Nitrogen Breakthrough:** The boiling point of liquid Nitrogen is 77 K (−196° C), while liquid Helium boils at 4.2 K. Because $T_c = 93$ K > 77 K, YBCO can be cooled using inexpensive, non-toxic liquid nitrogen ($\approx 100\times$ cheaper than liquid helium), enabling practical industrial applications.

PART B: MAGNETIC MATERIALS

5. Origin of Magnetism and Magnetic Parameters

Magnetism arises fundamentally from the continuous orbital motion and intrinsic quantum mechanical spin of electrons within atoms.

5.1 Magnetic Moments in Atoms

1. **Orbital Magnetic Dipole Moment (μ_L):** Arises from the orbital revolution of electrons around the positive nucleus (analogous to a microscopic current loop):
$$\mu_L = -\frac{e}{2m_e}\boldsymbol{L}$$
2. **Spin Magnetic Dipole Moment (μ_S):** Arises from the intrinsic quantum mechanical spin angular momentum of the electron:
$$\mu_S = -\frac{e}{m_e}\boldsymbol{S}$$
3. **The Bohr Magneton (μ_B):** The fundamental quantum mechanical unit of atomic magnetic dipole moment:
$$\boldsymbol{\mu_B = \frac{eh}{4\pi m_e} = 9.274 \times 10^{-24}\,\text{J T}^{-1} \quad (\text{or A m}^2)}$$

5.2 Fundamental Magnetic Quantities and Relationships

- **Magnetic Induction / Flux Density (B):** Total magnetic flux per unit area: Tesla (T) = Wb m^{-2}.
- **Magnetic Field Strength (H):** External magnetizing field: A m^{-1} or Oersted.
- **Intensity of Magnetization (M):** Magnetic dipole moment induced per unit volume:

$$M = \frac{\text{Total Magnetic Moment}}{\text{Volume}} \quad (\text{Units: A m}^{-1})$$

- **Constitutive Equation of Magnetism:**

$$\boldsymbol{B} = \boldsymbol{\mu_0}(\boldsymbol{H} + \boldsymbol{M})$$

where $\mu_0 = 4\pi \times 10^{-7}$ T m A^{-1} (H m^{-1}) is the permeability of free space.
- **Magnetic Susceptibility (χ):** Measure of how readily a material is magnetized in an external field:

$$\boldsymbol{\chi} = \frac{\boldsymbol{M}}{\boldsymbol{H}} \quad (\text{Dimensionless})$$

- **Relative Permeability (μ_r):** Ratio of material permeability (μ) to free space permeability (μ_0):

$$\boldsymbol{\mu_r} = \frac{\boldsymbol{\mu}}{\boldsymbol{\mu_0}} = \mathbf{1} + \boldsymbol{\chi}$$

6. Classification of the Five Types of Magnetic Materials

Material Type	Spin Alignment	Susceptibility (χ)	Permeability (μ_r)	Representative Examples
Diamagnetic	All spins paired (↑↓↑↓)	Small & negative (-10^{-5})	Slightly < 1 (0.99999)	Bismuth, Copper, Gold, Water, $NaCl$
Paramagnetic	Random unpaired (↑↗↓↖)	Small & positive ($+10^{-4}$ to $+10^{-2}$)	Slightly > 1 (1.0001)	Aluminium, Platinum, O_2, $CuSO_4$
Ferromagnetic	Parallel domain spins (↑↑↑↑)	Enormous & positive ($10^3 - 10^6$)	Extremely large ($\mu_r \gg 1000$)	Iron (Fe), Cobalt (Co), Nickel (Ni), $Nd_2Fe_{14}B$
Antiferromagnetic	Antiparallel equal (↑↓↑↓)	Small & positive (10^{-4} to 10^{-2})	Slightly > 1	MnO, FeO, CoO, NiO, Cr
Ferrimagnetic	Antiparallel unequal (↑↑↓↑↑)	Large & positive ($10 - 10^4$)	Large ($\mu_r \gg 1$)	Magnetite (Fe_3O_4), Ferrites ($NiFe_2O_4$)

6.1 Temperature Dependence: Curie and Néel Temperatures

1. **Curie's Law (Paramagnetism):** Thermal agitation disrupts dipole alignment:

$$\chi = \frac{C}{T} \quad (C = \text{Curie Constant})$$

2. **Curie-Weiss Law (Ferromagnetism):** Above the **Curie Temperature (T_C)**, thermal vibrations destroy spontaneous domain alignment, transforming the material into a normal paramagnetic state:

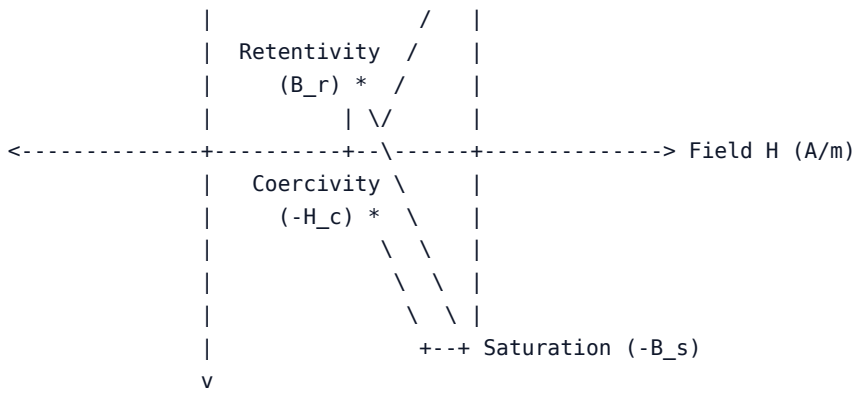
$$\chi = \frac{C}{T - \theta} \quad (\text{for } T > T_C)$$

   - Values: Iron $T_C = 770°$ C (1043 K); Cobalt $T_C = 1120°$ C (1393 K); Nickel $T_C = 358°$ C (631 K).
3. **Néel Temperature (T_N) (Antiferromagnetism):** Susceptibility increases with temperature up to T_N as thermal energy overcomes antiparallel exchange coupling, peaking at T_N, then decreases as a paramagnet above T_N.

7. Magnetic Hysteresis Loop ($B - H$ Curve) and Data Storage

When a ferromagnetic material is subjected to a cyclic magnetizing field (H), the flux density (B) lags behind H. This phenomenon is called **Magnetic Hysteresis** (from Greek **hysterein**, "to lag behind").





- **Key Hysteresis Parameters:**
  1. **Saturation Magnetization (B_s):** The maximum flux density attained when all magnetic domains are fully rotated and aligned parallel to external field H.
  2. **Retentivity / Remanence (B_r):** The residual magnetic flux density remaining in the core when the external magnetizing field is reduced to zero ($H = 0$).
  3. **Coercivity / Coercive Force (H_c):** The reverse magnetic field strength required to completely demagnetize the material and reduce residual flux to zero ($B = 0$).
  4. **Hysteresis Energy Loss:** The energy dissipated as heat per unit volume per cycle is directly proportional to the **enclosed area** of the hysteresis loop.

7.1 Soft vs. Hard Magnetic Materials

Parameter	Soft Magnetic Materials	Hard Magnetic Materials
Hysteresis Loop Area	**Narrow / Slim loop:** Extremely low hysteresis energy loss.	**Broad / Wide loop:** Large enclosed loop area.
Coercivity (H_c)	**Low coercivity:** $H_c < 100 \text{A m}^{-1}$ (easily magnetized and demagnetized).	**High coercivity:** $H_c > 10^4 - 10^6 \text{A m}^{-1}$ (resists demagnetization).
Permeability (μ_r)	**High permeability:** Easily channels magnetic flux lines.	**Low permeability:** Requires strong fields to magnetize.
Primary Composition	Silicon steel ($4\% Si-Fe$), Permalloy ($78\% Ni-Fe$), Soft Ferrites.	Alnico ($Al-Ni-Co$), Hard Ferrites ($BaFe_{12}O_{19}$), Neodymium-Iron-Boron ($Nd_2Fe_{14}B$).
Applications	Transformer laminated cores, electric motor stators, inductors, magnetic shields.	Permanent magnets, DC motors, loudspeakers, magnetic door latches.

7.2 Applications of Magnetic Materials in Digital Data Storage

1. **Magnetic Hard Disk Drives (HDDs):**
   - **Storage Platter:** Aluminium or glass substrate coated with an ultrathin ($\approx 10-20$ nm) hard magnetic film of Cobalt-Chromium-Platinum alloy ($CoCrPt$). Tiny localized magnetic domains oriented perpendicular to platter surface represent binary bits (0 and 1).
   - **Read-Write Heads:** Employ **Giant Magnetoresistance (GMR)** or **Tunneling Magnetoresistance (TMR)** spin-valve sensors consisting of alternating ferromagnetic and non-magnetic layers. When the head passes over a magnetized bit on the rotating disk, the local magnetic field rotates magnetization in the free layer, changing electrical resistance by $20-70\%$ to read data.
2. **Magnetic Random Access Memory (MRAM):**
   - Non-volatile solid-state memory utilizing **Magnetic Tunnel Junctions (MTJ)**.

- An MTJ consists of two ferromagnetic layers separated by an insulating magnesium oxide (MgO) barrier (1 nm). When magnetizations are parallel, quantum tunneling resistance is low (binary 0); when antiparallel, resistance is high (binary 1).
- Retains memory permanently without power; immune to ionizing radiation.